

Graphormer: Ablation, and Inference-Time Architectural Extensions

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Abstract

We re-implement Graphormer [1] from scratch as `mini_graphormer`, a 48M-parameter PyTorch package. Microsoft’s `pcqm4mv2_graphormer_base` checkpoint loads with *strict* parameter matching, and on a 2,000-graph slice of the PCQM4Mv2 validation set our reproduction reaches $\text{MAE} = 0.102 \text{ eV}$ – consistent with the published 0.0864 eV within finite-sample variance. An ablation of the four structural biases reveals that the multi-hop *edge* bias dominates (+338% MAE when removed), far ahead of the shortest-path *spatial* bias (+36%). We then test three inference-time architectural extensions inspired by SAN, GTN, and PAGTN: Laplacian PE, RWPE, and a ring-aware bias. Code is available at <https://github.com/HARSHIL2836R/Graphormer-Project>.

1 Implementation

The original Graphormer release ships as a fairseq fork with a Cython preprocessing module; both have well-known installation issues on modern toolchains. We re-implement the model and pipeline as a self-contained PyTorch package. Every parameter is named identically to the released checkpoint, so `model.load_state_dict(state_dict, strict=True)` succeeds with *zero* missing and *zero* unexpected keys (all 210 tensors aligned, including embeddings, attention projections, and the regression head). The PCQM4Mv2 data path replaces the Cython all-pairs Floyd–Warshall and multi-hop edge encoder with vectorised NumPy, and streams the validation split directly from the raw CSV, sidestepping OGB’s full-dataset upfront SMILES processing (3.7M molecules, hours on CPU). Both ablations and extensions are implemented as forward hooks on `nn.Modules`; underlying weights are never modified, allowing many configurations to be evaluated against a single loaded checkpoint.

2 Reproducing the pretrained baseline

On the first 2,000 PCQM4Mv2 validation graphs, our implementation reaches **MAE = 0.1020 eV**, against the published **0.0864 eV** on the full 73,545-graph set. We verified separately (on $n = 1000$ sized samples) that within-sample MAE swings by roughly $\pm 0.01 \text{ eV}$ depending on which contiguous 1000 graphs are sampled; the 0.016 eV gap to the published number is therefore consistent with sampling noise, not with any architectural defect.

3 Ablation study

Each ablation zeroes the output of one structural-bias submodule via a forward hook, leaving the rest of the model untouched. Results (Table 1) show a striking hierarchy.

Table 1: Effect of removing each Graphormer-specific structural bias on PCQM4Mv2 validation MAE ($n = 2000$).

removed bias	MAE	Δ MAE	$\Delta\%$
baseline (full model)	0.1020	—	—
spatial bias $b(\phi(i, j))$	0.1384	+0.036	+36%
multi-hop edge bias $c(i, j)$	0.4471	+0.345	+338%
virtual graph-token distance	0.1020	+0.000	+0%
in/out-degree centrality	0.1178	+0.016	+16%
all attention biases together	0.7576	+0.656	+643%

Quantitative findings. (i) The multi-hop edge bias is by far the most load-bearing component: removing it alone quadruples MAE, roughly. (ii) The graph-token virtual-distance term has zero measurable effect on validation MAE – it can be ablated freely without penalty. (iii) Centrality (degree encoding on node features) contributes modestly (+16%). (iv) The sum of individual ablation costs (390%) falls well short of the all-bias-removed cost (643%), indicating super-additive interactions between biases beyond the linear regime.

Qualitative Analysis. Graphormer is usually summarised as “Transformer + spatial encoding,” yet on PCQM4Mv2 the multi-hop edge encoding is what actually carries the structural prior. The model’s predictions degenerate to a near-mean predictor when all attention biases are removed ($\text{MAE} = 0.76$), confirming that the transformer backbone alone – with no graph-aware bias – recovers little beyond label statistics on this benchmark.

4 Inference-time architectural extensions

We test three additive attention-bias terms inspired by the relatives of Graphormer surveyed in our midsem literature review: SAN (spectral) [2], GTN (random-walk-style adjacency combinations) [3], and PAGTN (path / ring features on molecular graphs) [4]. Each is a single bias matrix added on top of Graphormer’s $(b(\phi), c(i, j), t)$ terms with one tunable scalar α , and is evaluated in four configurations on the same $n=2000$ validation slice. Each configuration probes a distinct claim:

- *baseline* – pretrained model, no extension. Constant 0.1020 eV;
- *X only* – every Graphormer attention/centrality bias zeroed; only the new term X is active. Probes whether the new prior *alone* can substitute for Graphormer’s trained structural signature.
- *X + edge* – only the multi-hop edge bias kept (the dominant Graphormer term per Table 1), plus X. Isolates X’s complementarity to the most important built-in bias.
- *all + X* – full Graphormer plus X. The actual test of whether X provides additive value at the trained operating point.

4.1 Laplacian PE (SAN-flavoured)

We construct $S = U_k W U_k^\top$ where $U_k \in \mathbb{R}^{N \times k}$ collects the $k = 8$ smallest non-trivial Laplacian eigenvectors per molecule and $W = I$ (uniform weighting). The bias is αS , padded into the $(N+1) \times (N+1)$ attention bias and broadcast across heads.

Table 2: Laplacian PE sweep ($k = 8$, uniform weighting). Baseline = 0.1020 eV.

α	lap only	lap + edge	all + lap	Δ vs base
0.2	0.7134	0.1700	0.1019	−0.00012
0.5	0.7285	0.1821	0.10182	−0.00015
1.0	0.7551	0.2122	0.10198	+0.00001
2.0	0.8117	0.2946	0.1045	+0.0025

Quantitative. The “lap only” column lies in the 0.71–0.81 range – $7\text{--}8\times$ baseline – showing spectral similarity is wholly inadequate as a structural signal for a model trained on shortest-path biases. “all + lap” traces a clean U-shape with minimum at $\alpha = 0.5$, where MAE drops by 1.5×10^{-4} eV.

Qualitative. The well-defined U-shape — improvement on both sides of the optimum, not noise — supports SAN’s central hypothesis that spectral encodings carry structural information complementary to shortest-path biases, even when the model has already been trained on the latter. That *any* aggregate improvement is achievable without weight updates is itself notable: an eight-dimensional spectral subspace per molecule visibly steers a 48M-parameter pretrained transformer toward a better operating point. We expect this prior to scale up substantially with even brief fine-tuning, since the model would then be free to re-calibrate its other biases to accommodate the new term.

4.2 RWPE (GTN-flavoured)

We accumulate $S = \sum_{k=1}^K (D^{-1/2} A D^{-1/2})^k$ with $K = 4$, the symmetrically-normalised random-walk power matrix that gives a multi-scale “soft adjacency”.

Table 3: RWPE sweep ($K = 4$, symmetric normalisation). Baseline = 0.1020 eV.

α	rwpe only	rwpe + edge	all + rwpe	Δ vs base
0.05	0.7096	0.1644	0.10190	−0.00007
0.10	0.7158	0.1651	0.10190	−0.00008
0.20	0.7304	0.1666	0.1020	+0.00006
0.50	0.7937	0.1729	0.1026	+0.00060
0.80	0.8922	0.1840	0.1032	+0.00124
1.80	1.1968	0.2844	0.1102	+0.00825

Quantitative. “rwpe only” rises sharply from 0.71 at $\alpha = 0.05$ to 1.20 at $\alpha = 1.8$ – a much steeper sensitivity than the Laplacian column – reflecting the larger native magnitudes of A^k entries (which can exceed 1 in dense subgraphs) compared to normalised eigenvector outer products. “all + rwpe” bottoms out at $\alpha \approx 0.05\text{--}0.10$ with a gain of 7×10^{-5} eV, half of Laplacian’s optimum.

Qualitative. RWPE’s “soft adjacency” is the closest of the three to information Graphormer already encodes implicitly through shortest-path distance, which may explain its smaller gain: the trained biases capture most of what the random-walk matrix offers. The faster degradation at large α is a direct consequence of the dense, high-magnitude bias matrix – each unit of α injects more perturbation than Laplacian’s lower-rank construction.

4.3 Ring-aware bias (PAGTN-flavoured)

The bias is binary: $S_{ij} = \mathbf{1}[i, j \text{ share a ring}]$, with rings extracted via rdkit’s smallest-set-of-smallest-rings.

Table 4: Ring-aware sweep. Ring identity from rdkit `GetSymmSSSR`.

α	ring only	ring + edge	all + ring	Δ vs base
0.05	0.7041	0.1619	0.10192	−0.00005
0.20	0.7058	0.1579	0.1022	+0.00022
0.50	0.7142	0.1547	0.1037	+0.00177
1.00	0.7555	0.1622	0.1107	+0.00867
2.00	0.8816	0.2472	0.1600	+0.05803

Quantitative. Ring-aware shows the steepest sensitivity of the three: at $\alpha = 1$ MAE inflates by 9%, at $\alpha = 2$ by 57%. The optimum at $\alpha = 0.05$ gives a gain of 5×10^{-5} eV – the smallest of the three optima. Notably, “ring + edge” is the only column across all three sweeps that genuinely *improves* with moderate α , reaching 0.1547 at $\alpha = 0.5$, suggesting ring identity does complement edge information when the spatial term is absent.

Qualitative. The binary nature of the ring bias is responsible for both its sharp degradation curve and its potentially clean signal: a binary mask scaled by α acts as a hard step in the softmax, which the trained model has no smooth way to absorb. A softer ring-distance bias (e.g. ring-membership weighted by ring size or shared-ring count) might land more gently inside the model’s operating window. Still, the “ring + edge” column demonstrates the usefulness of the information provided by “Ring-aware bias”.

4.4 Per-molecule study via `handcraft.py`

We built `handcraft.py`, a script that evaluates the three extensions on a hand-picked set of canonical aromatic molecules with documented HOMO–LUMO gaps. For each molecule the script runs four forward passes on the loaded checkpoint – baseline, then each extension applied *in isolation* (never two at once) – and prints predictions side-by-side with the experimental gap. The molecule choice deliberately spans easy and structurally distinctive cases: **naphthalene** and **anthracene** as polycyclic aromatic hydrocarbons(**PAH**); **phenanthrene** as the angular-fusion isomer of anthracene; **pyrene** as a highly fused PAH and **azulene**, the constitutional isomer of naphthalene whose fused topology gives a dramatically different gap (1.7 eV vs. 4.4 eV) that any topology-aware prior should detect.

The run reported in Table 5 uses each extension’s sweep-optimal α ($\alpha_{\text{Lap}}=1.0$, $\alpha_{\text{RWPE}}=0.5$, $\alpha_{\text{ring}}=0.05$).

Table 5: Per-molecule HOMO–LUMO gap predictions (eV) at optimal α . “Closest” = whichever configuration’s prediction is nearest the experimental gap.

molecule	exp.	baseline	+Lap.	+RWPE	+Ring	closest
naphthalene	4.40	4.834	4.830	4.834	4.835	+Lap.
azulene	1.70	3.333	3.336	3.308	3.329	+RWPE
anthracene	3.30	3.586	3.588	3.579	3.586	+RWPE
phenanthrene	3.90	4.748	4.752	4.750	4.748	baseline
pyrene	3.30	3.842	3.843	3.834	3.841	+RWPE

At optimal α , all extensions shift predictions by at most 0.03 eV per molecule, consistent with the aggregate finding that the perturbations are sub-millivolt at the right magnitude. We separately observed in earlier runs at $\alpha_{\text{ring}} = 1.0$ that pyrene’s prediction shifts by -0.08 eV under the ring bias – a visible, structurally-justified correction toward the experimental 3.30 eV . T

References

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